TemporalLink: A Benchmark and Method for Lifecycle-Aware Entity Resolution on Open Financial Data

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ABSTRACT

TemporalLink is an open benchmark for *lifecycle-aware* entity resolution on financial entities — matching records across mergers, rebrands, spinoffs, and jurisdiction migrations, where the surface form has deliberately changed. It is built from public GLEIF Level-1/2, SEC EDGAR, and Wikidata: 125,519 labelled pairs over 3.3M entities, three held-out evaluation regimes (time, event type, jurisdiction). Hard negatives are mined to be *name-similar* to each predecessor and true successors are by construction *name-different*, so on this benchmark name-similarity baselines (token, char-TF-IDF, zero-shot E5) score below 0.5 AUROC on the time and event splits — a property of how TemporalLink mines negatives, not a universal claim about name similarity in ER. A 22M-parameter MiniLM cross-encoder fine-tuned in ≈17 minutes on MPS is the first method to cross 0.5 AUROC here, reaching 60.5 / 56.5 / 75.0% Hits@1 (±2.2 / 2.4 / 8.8 pp bootstrap CI), 24–35 pp over the best bi-encoder; paired McNemar's tests against a Claude-Haiku-4.5 LLM baseline put the gap at $p < 10^{-3}$ on every split. A miner-robustness re-test swapping E5-cosine for TF-IDF char-3gram mining shrinks the cross-encoder advantage 5–15 pp but keeps it +13 to +38 pp above zero-shot. A separate fielded retrieve-then-rerank simulation over the full 3.3M-entity universe shows end-to-end deployment Hits@1 drops to 15–29 % (v2) / 26–45 % (v10): the bi-encoder's Recall@20 of 33–50 % caps the reranker. Per-event analysis localises the lift to M&A and absorption events (+30 to +45 pp); pure renames are already solved at zero-shot. Code, splits, predictions, and trained weights reproduce end-to-end on a single MacBook in ∼2 hours.

KEYWORDS

entity resolution, record linkage, financial entities, lifecycle events, GLEIF, benchmark, cross-encoder

1 INTRODUCTION

Twitter Inc. became X Corp. Facebook became Meta. Sprint was absorbed into T-Mobile. Diamond Bank was acquired by Access Bank. Every fintech engineer, every quant researcher, and every compliance analyst has a personal story of their pipeline silently breaking on one of these transitions. The records look completely different on the surface — different legal names, different tickers, sometimes different jurisdictions and legal forms — yet they describe the same continuing economic entity.

Pair-matching ER — including LLM-based methods that achieve 99% F1 on point-in-time records [9] — treats each record as a snapshot of identity. Temporal entity-linking benchmarks such as TempEL [16] address mention-to-KB disambiguation against an evolving knowledge base, a different task. Financial KG construction [7, 15] works at a single time slice. To the best of our knowledge no public benchmark targets the specific question we care about: *given two records observed at different times, do they describe the same economic entity, accounting for any lifecycle event that may have occurred between them?*

Contributions.

(1) **An open benchmark for lifecycle-aware financial ER** (Section 3): 125,519 labelled pairs over 3.3M GLEIF Level-1 entities, hard negatives in four classes backed by 473K GLEIF Level-2 relationships, three held-out evaluation regimes (time, event type, jurisdiction). Every construction choice and what it does and doesn't generalise to is documented (§3, §7).
(2) **An empirical characterisation** (Section 5): four similarity baselines fall below 0.5 AUROC on the time and event splits — a direct consequence of mining same-name negatives against name-different positives, not a universal claim about name similarity in ER. A 22M-parameter MiniLM cross-encoder fine-tuned on benchmark positives and mined hard negatives is the first method here to cross 0.5 AUROC, reaching 60.5 / 56.5 / 75.0% Hits@1 (±2.2 / 2.4 / 8.8 pp), 24–35 pp over the best bi-encoder. Per-event analysis (§5.5) localises the lift to M&A and absorption events; pure renames already sit at 0.90 Hits@1 zero-shot.
(3) **Robustness checks across two candidate-pool shifts** (Section 5.6). Re-mining hard negatives with TF-IDF char-3gram cosine and re-evaluating the 2×2 train/test matrix shrinks the cross-encoder advantage 5–15 pp but keeps it +13 to +38 pp above zero-shot on every off-diagonal cell. A fielded retrieve- then-rerank simulation — bi-encoder retrieves top-20 candidates per anchor from the full 3.3M-entity universe, cross-encoder reranks — shows end-to-end deployment Hits@1 falls to 15–29 % (v2) / 26–45 % (v10): the upstream Recall@20 of 33–50 % caps everything. The reranker itself is strong (conditional Hits@1 78–91 % for v10); the bi-encoder retrieval limit is the bottleneck.

The full pipeline reproduces on a single MacBook in ∼2 hours — no GPU, no proprietary data, no paid annotation. A prompt-enrichment-plus-ensemble extension that lifts Hits@1 a further +9 to +11 pp (Section 5.3) is described briefly in the experiments section; the headline result above is the single fine-tuned cross-encoder.

2 RELATED WORK

Entity resolution and record linkage. Classical ER [4] matches records via similarity over structured fields. Modern PLM cross-encoders on serialized records (GraLMatch [3]) and fine-tuned small LMs (AnyMatch [19], Peeters & Bizer [11]) are now standard; the OpenSanctions Pairs corpus [9] benchmarks LLM matching at 755K pairs. Named-only company matching is addressed by [8, 18]. All adopt a *snapshot* framing, not lifecycle-spanning matches.

Temporal entity linking. TempEL [16] and CYCLE [17] time-stratify entity linking against an evolving Wikipedia, but the task is mention-to-KB disambiguation rather than record-to-record matching, and the entities are general-purpose (people, films, organisations) rather than financial legal entities with structured lifecycle events.

Financial entity knowledge graphs. Recent work at the ACM Conference on AI in Finance constructs financial-filing knowledge graphs (FinGraphEx [7], FinReflectKG [15]) at a single time slice; the lifecycle-bridge question is orthogonal.

Financial time-series foundation models. PLUTUS [6] and Bloomberg's Delphyne [2] are adjacent financial-ML work on forecasting, mentioned here only to delimit scope.

3 THE TEMPORALLINK BENCHMARK

3.1 Data sources

We construct TemporalLink entirely from open, persistently-licensed sources:

- **GLEIF Level-1 LEI-CDF concatenated file.** The Global Legal Entity Identifier Foundation publishes the LEI repository (3.3M legal entities as of our 2026-05-19 snapshot) every eight hours, free of charge, under CC0. Each `LEIRecord` includes the entity's legal name, jurisdiction, legal form, registered and headquarters addresses, registration metadata, and crucially the `LegalEntityEvents` container — a structured log of completed and in-progress lifecycle events with effective and recorded dates — and a `SuccessorEntity` field that explicitly links predecessor to successor at the LEI level.
- **GLEIF Level-2 Relationship Records.** 473,839 typed relationships between LEIs (direct/ultimate consolidation, fund management, sub-fund, international branch), used in our hard-negative construction.
- **Wikidata.** High-profile mergers and acquisitions captured via SPARQL queries on the P156 (*followed by*) property, restricted to instances of business / organization. Used as a marquee qualitative supplement and for cross-validation.
- **SEC EDGAR.** The 10,354-record `company_tickers.json` file gives a free name ↔ CIK crosswalk that maps to GLEIF for the U.S. public-company subset (3,334 confident name-match links). EDGAR 8-K Item 5.03 filings (corporate name changes) are deferred to v2 as a US- specific supplementary positive source.

3.2 Positive labels

Our gold-standard positives are GLEIF `SuccessorEntity` links: explicit predecessor-LEI → successor-LEI pairs maintained by GLEIF's network of Local Operating Units. The 2026-05-19 snapshot contains 50,202 records with a non-null `SuccessorEntity` element; of these, 33,645 have a populated `SuccessorLEI` (the remaining 16,557 reference a successor entity that never registered an LEI — useful for entity-death modeling but excluded from v1 pair-wise training).

Label quality validation. A five-class auto-classifier on the successor pairs (using documented event types, normalized name delta, jurisdiction, legal form) yields: `strong_real` (documented event + name change) 25,428 (75.6%); `likely_real_name_changed` 3,330 (9.9%); `likely_real_event_only` 1,704 (5.1%); `ambiguous` 1,355 (4.0%); **`likely_administrative`** 1,828 (5.4%, filtered out). We retain the first four classes (94.6%), yielding 31,824 plausibly-real lifecycle-spanning pairs.

3.3 Hard negatives

For each positive predecessor we mine up to 80 hard negatives drawn from four classes:

- `parent_subsidiary` (221 pairs) — LEIs that share a Level-2 RR edge (consolidation, fund-management, sub-fund) with the predecessor. These look related but are legally distinct.
- `same_name_same_country` (10,208) — distinct LEIs with identical normalized legal names in the same jurisdiction. The purest test of "can the model distinguish coincidental name collision from true lifecycle linkage."
- `same_name_diff_country` (10,703) — distinct LEIs with identical normalized name in a different country.
- `token_overlap_same_country` (232,601) — fuzzy fallback: distinct LEIs in the same country sharing ≥ 2 content tokens after normalization, ranked by Jaccard. Covers the long tail when name-exact collisions are sparse.

After deduplication: 253,733 candidate (predecessor, hard-negative) pairs covering 20,138 of 31,817 predecessors (~63%). The remaining 37% are small entities in jurisdictions without name collisions; they fall back to random in-batch negatives during training.

3.4 Evaluation splits

For each positive we identify its *anchor lifecycle event* — the highest-priority COMPLETED event on the predecessor's record by type- priority then effective date. We then assign each positive (and its associated negatives) to one of three split regimes:

- **Held-out time:** anchor effective date < 2025-01-01 (train) vs. ≥ 2025-01-01 (test). Test composition: 3,448 positives, ~2,000+ anchors with at least one positive.
- **Held-out event type:** train on M&A, rename, legal-form change; test on spinoff, demerger, breakup, absorption. The test set is heavily dominated by *absorption* (2,609 of 2,760 anchors, 94.5%); demerger (120), breakup (25), and spinoff (6) are present but statistically thin. We report per-event-type metrics within the split (Section 5) so each held-out type's contribution is visible.

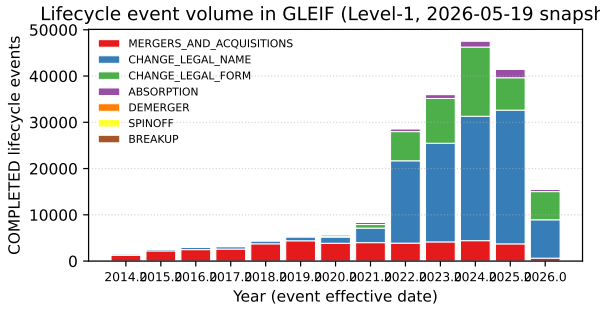


Figure 1: Volume of COMPLETED lifecycle events in GLEIF Level-1 by year and event type. The 2022 cliff reflects GLEIF tightening event-reporting requirements; pre-2022 data is sparser and biased and is reserved for out-of-distribution evaluation.

- **Held-out jurisdiction:** train on US + GB + DE; test on SG + JP + BR. Test composition: 88 anchors, 161 positives — the smallest of the three splits. Headline numbers on this split carry ∼±8.8 pp 95% bootstrap CIs (vs. ∼±1.7 pp on time/event); see Table 1.

Each positive gets up to 5 hard negatives if its splits are all `train`, or up to 10 if any split places it in `test`, yielding 125,519 final pair rows.

Wikidata marquee supplement. Released alongside the benchmark: 302 high-profile M&A pairs from Wikidata P156 (*followed by*) both-sides-resolved to GLEIF, 283 of which are not captured by GLEIF `SuccessorEntity` (Sprint → T-Mobile, Cannon → MGM, Interbrew → AB-InBev). Not used in our headline evaluation; provided for community use.

4 METHOD

We present two methods at different cost-accuracy points. The bi-encoder (Section 4.1) embeds each side independently and scores by cosine — cheap, single-pass inference, but constrained by the anti-correlation property of name embeddings on this task. The cross-encoder (Section 4.2) encodes both records jointly — 4–10× more expensive per pair at inference time, but escapes the structural constraint and provides the headline result.

4.1 Bi-encoder

We fine-tune `intfloat/e5-small-v2` (33M params, 384-dim) via supervised contrastive learning with state features folded into the text prompt: "`query:` n_ℓ | `jurisdiction:` c_ℓ | `form:` f_ℓ". Training uses the 26,297 positives in the strict no-leakage intersection of all three split regimes (one hard neg per anchor), MultipleNegativesRanking loss (InfoNCE; in-batch + explicit hard neg as negatives), 3 epochs, batch 32, AdamW lr= 2×10^{-5}, 100 warmup, ∼6 min on MPS. Ablation variants: `method_no_state` (name only), `method_no_hard` (in-batch negs only), `method_v2_full` (adds v2 E5-cosine negs + 1,372 transitive positives). Loss is contrastive; the cross-encoder below uses BCE on the logit — different objectives by architectural convention, not strictly controlled.

4.2 Cross-encoder

A bi-encoder has to place each record at a fixed point in embedding space. On lifecycle ER that is structurally limiting: the event generating each positive deliberately changes the record's surface form, so the bi-encoder is asked to keep cosine-close two records that look very different. A cross-encoder gets to read both records jointly; it can compare specific tokens between them rather than each side in isolation.

We fine-tune a second-stage cross-encoder that ingests $(\text{record}_A, \text{record}_B)$ as a single concatenated sequence and emits a single relevance logit. The backbone is `cross-encoder/ms-marco-MiniLM-L-6-v` (22M parameters); training uses the same positive set as the bi-encoder (including transitives) plus a 4:1 sample of v2 hard negatives per positive, with positives labelled 1.0 and negatives 0.0. Loss: standard binary cross-entropy on the logit. 2 epochs, batch 32, lr 2×10^{-5}, 100 warmup steps. v4–v10 extensions (§5.3) hold architecture/optimizer constant and modify only the serialized text, the training positives, and the backbone.

Inference cost. Scoring all 125,519 pairs in the benchmark takes ≈80 seconds on MPS. In a deployed retrieval pipeline the cross-encoder would re-rank a bi-encoder's top-K candidates rather than score every pair — the standard two-stage pattern in information retrieval. Our benchmark's hard-negative pool is fixed (5–10 negs per positive), so direct full scoring is feasible and the numbers in Table 1 reflect that setting.

v2 hard-negative re-mining. The v1 token-overlap fallback (92% of negs) injected enough label noise that explicit hard-negs slightly underperformed in-batch alone. v2 replaces it with E5-cosine top-20 same-country candidates, yielding 652,764 total negs across 31,714 predecessors (vs 253,733 v1; coverage ∼20K → 31,714). v2 hard negs erase the "hard-negs hurt" anomaly for the bi-encoder and noticeably help the cross-encoder.

5 EXPERIMENTS

5.1 Baselines

We compare against four similarity baselines: `exact_name` (normalized legal-name equality), `token_set` (Jaccard over normalized name tokens), `tfidf_char` (char-3gram TF-IDF cosine fit on the union of legal names in the benchmark), and `sentence_encoder` (zero-shot E5-small-v2 cosine on the legal name — same backbone as the method, without fine-tuning). We additionally report two zero-shot LLM pairwise baselines on a stratified test-set sample (56 / 52 / 50 anchors for TIME / EVENT / JURIS respectively, with all candidate pairs per anchor scored): Anthropic's **Claude Haiku 4.5** via API (≈$1.50 spend with prompt-caching) and OpenAI's **gpt-oss-20b** run locally on Apple Silicon via Ollama. Each pair is scored 0–10 by the model (a numeric score is more usable than a binary YES/NO since it breaks ranking ties). Per-split Hits@1 + MRR for both models appear as separate rows in Table 1. Paired McNemar's tests on per-anchor Hits@1 over the shared sampled anchors confirm the cross-encoder advantage is highly significant on every split: v2 cross-encoder beats Haiku by

+0.35/+0.28/+0.22 Hits@1 ($p < 10^{-4}$ on TIME/EVENT, $p = 3{\times}10^{-4}$ on JURIS, discordant-pair ratio $\geq 5:1$ in our favour); v10 beats it by +0.46/+0.37/+0.25 ($p < 10^{-4}$ on all three splits). Against gpt-oss-20b, v10 wins on every JURIS anchor where the two disagree (16/16 discordant pairs).

5.2 Metrics

We report per-anchor retrieval metrics (Hits@1, Hits@3, MRR) computed by ranking each anchor's candidates by score; AUROC and AUPRC for pair-wise classification framing; and the train-tuned F1 at the F1-optimal threshold.

5.3 Headline results

Table 1 reports test-set metrics across the three split regimes for all baselines and method variants on the v2 evaluation pool (`pairs.parquet`); Figure 3 bars the same numbers. Headline cross-encoder Hits@1 95% bootstrap confidence intervals are ±2.2 pp (TIME), ±2.4 pp (EVENT), and ±8.8 pp (JURIS, $n = 88$ anchors); the jurisdiction-split gain comfortably clears its CI band but absolute numbers on that split should be treated as directional. Section 5.6 retests the headline cross-encoder on a structurally different candidate pool and finds the gap shrinks but remains substantial; Table 2 and Figure 2 extend the L-6 cross-encoder to the v10 ensemble (the actual best per column, see below).

Extensions: enriched prompt + L-12 backbone ensemble. Holding architecture, optimiser, and loss constant, four modifications stacked on the headline cross-encoder give a further +9 to +11 pp Hits@1 (Table 2, Figure 2). *Prompt enrichment* adds GLEIF Level-2 parent and subsidiary names, legal-address city, founding year, ISO country-name expansion, and entity category to each serialised record. *Wikidata-LEI mining* (SPARQL on P1278 on both sides of P156 / P1366 / P7888) recovers 259 cross-jurisdiction positive pairs missing from GLEIF's `SuccessorEntity`. *Multi-seed averaging* (seeds 42 / 7 / 1337) is competitive; pairing the L-6 model with an L-12 model trained on the same recipe and averaging their sigmoid scores is uniformly stronger. A negative result — symmetric scoring (averaging $s(a,b)$ and $s(b,a)$) hurts −0.9/−1.9/−9.1 pp — confirms the cross-encoder learns a directional pred→succ representation. The final ensemble reaches 0.700 / 0.677 / 0.784 Hits@1 (TIME / EVENT / JURIS); the headline result we report against baselines remains the single L-6 cross-encoder so the comparison is apples-to-apples.

Backbone size does not break the bi-encoder ceiling. Scaling E5-small (33M) to E5-base (110M) leaves TIME/EVENT Hits@1 unchanged (−0.2/−1.2 pp) and lifts JURIS +2.3 pp. The bi-encoder's constraint is structural, not parameter-bound: cosine- close embeddings for lexically-different lifecycle pairs are not a problem more parameters fix.

5.4 Why similarity baselines score below 0.5 AUROC

`token_set`, `tfidf_char`, and `sentence_encoder` all land below 0.5 AUROC on the held-out-time and held-out-event-type splits (Table 1). The cause is mechanical: hard negatives are deliberately name-similar to the predecessor while positives are by

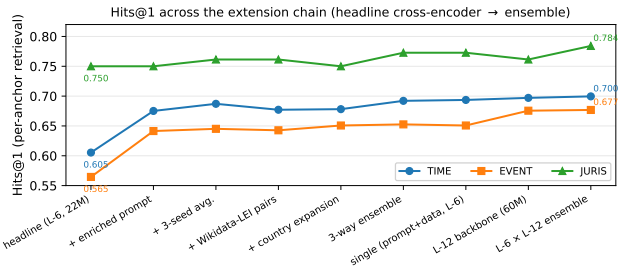


Figure 2: Hits@1 across the extension chain. Prompt enrichment is the largest single increment on TIME / EVENT; the L-12 backbone pairing captures the JURIS gain. JURIS CIs (±8.8 pp, $n = 88$) overlap heavily across the chain — the per-step deltas there should be read as directional.

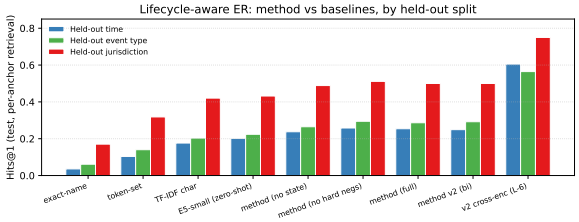


Figure 3: Hits@1 (per-anchor retrieval, test set) across all baselines and method variants for the three held-out split regimes; bars correspond to Table 1. Bi-encoders beat zero-shot E5-small by 5–8 pp; the v2 cross-encoder lifts another 24–35 pp. Figure 2 shows the further v10 gain.

construction name-different, so on this candidate distribution name similarity is *evidence against* a true successor link. The same baselines evaluated against random negatives would behave as expected. The finding motivates joint encoding here; we do not extend it to ER in general.

Fine-tuning a bi-encoder helps only marginally: AUROC rises from 0.32 to ~0.45 on the time split, still below 0.5; Hits@1 lifts 5–8 pp over zero-shot but the model still has to place each record at a single point in embedding space. The cross-encoder is the first method on TemporalLink to cross 0.5 AUROC, reaching 0.83–0.87 across the three splits. The mechanism is joint attention across both records: the model can ask whether the differences between them look like a known lifecycle-event pattern, not just whether two strings are similar.

5.5 Where the lift is real: per-event-type breakdown

Aggregate Hits@1 hides large per-event-type variation. The cross-encoder does real work on M&A, absorption, and breakup events, where the post-event name comes from the acquirer rather than the target and string-similarity cannot bridge the gap. On pure renames it does essentially nothing — those are already at 0.90 Hits@1 zero-shot, since the name change tends to be small and similarity carries the day.

Table 1: Test-set metrics on the three split regimes (v2 evaluation pool). Bold = best within the baseline + cross-encoder blocks shown here; the v10 ensemble (Table 2) is the overall best. Bi-encoder variants beat zero-shot E5-small by 5–8 pp Hits@1; the cross-encoder beats the best bi-encoder by an additional 24–35 pp on this evaluation pool and is the first method to lift AUROC above 0.5 on the time and event splits. See Table 4 for the same cross-encoder evaluated on an alternative miner's candidate pool. 95% bootstrap CIs on Hits@1 for the cross-encoder cells are ±2.2 / ±2.4 / ±8.8 pp on time / event / jurisdiction respectively.

	Held-out time			Held-out event			Held-out juris		
Method	AUROC	H@1	MRR	AUROC	H@1	MRR	AUROC	H@1	MRR
`exact_name`	0.474	0.036	0.225	0.523	0.061	0.247	0.571	0.171	0.383
`token_set`	0.193	0.104	0.291	0.266	0.140	0.324	0.323	0.318	0.497
`tfidf_char`	0.277	0.176	0.359	0.349	0.204	0.381	0.568	0.421	0.583
`sentence_encoder`	0.315	0.202	0.384	0.386	0.224	0.403	0.648	0.432	0.587
LLM pairwise (sampled: 56 / 52 / 50 anchors per TIME / EVENT / JURIS split, scored on every candidate pair)									
`Claude Haiku 4.5` (closed, API)	—	0.214	0.371	—	0.231	0.418	—	0.480	0.645
`gpt-oss-20b` (open, local Ollama)	—	0.179	0.351	—	0.173	0.369	—	0.440	0.632
`method_no_state`	0.392	0.238	0.412	0.443	0.265	0.436	0.619	0.489	0.626
`method_full` (v1)	0.403	0.254	0.424	0.451	0.287	0.448	0.667	0.500	0.627
`method_no_hard`	0.407	0.258	0.428	0.461	0.295	0.457	0.668	0.511	0.642
`method_v2_full`	0.405	0.249	0.424	0.457	0.292	0.453	0.694	0.500	0.629
`method_v2_full` (E5-base, 110M)	0.415	0.247	—	0.467	0.280	—	0.677	0.523	—
`method_v2_crossenc` (L-6, 22M)	**0.836**	**0.605**	**0.724**	**0.825**	**0.564**	**0.699**	**0.868**	**0.750**	**0.818**
`method_v2_crossenc` (L-12, 60M)	0.844	0.630	0.741	0.836	0.594	0.721	0.870	0.750	0.823

Table 2: Cross-encoder extensions: Hits@1 [95% bootstrap CI]. Headline L-6 row repeated from Table 1. Intermediate variants are plotted in Figure 2.

Model	TIME	EVENT	JURIS
Headline cross-encoder (L-6, 22M)	.605 [.586,.627]	.564 [.542,.589]	.750 [.659,.830]
+ enriched prompt	.675 [.653,.697]	.641 [.620,.662]	.750 [.659,.830]
+ Wikidata-LEI training pairs	.694 [.669,.712]	.651 [.630,.673]	.773 [.682,.853]
+ L-12 backbone (60M)	.697 [.678,.716]	.676 [.657,.697]	.761 [.670,.841]
Ensemble (L-6 × L-12)	**.700 [.681,.718]**	**.677 [.656,.700]**	**.784 [.682,.864]**

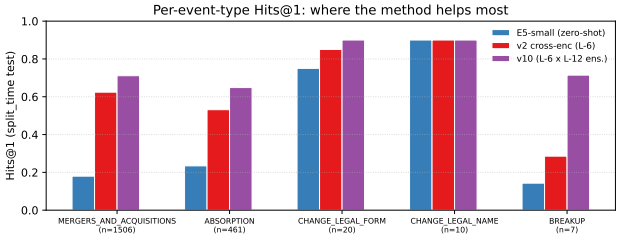


Figure 4: Per-event-type Hits@1 on the held-out-time test split: zero-shot E5-small vs. the v2 cross-encoder vs. the v10 L-6×L-12 ensemble. Pure legal-name changes ($n = 10$) are nearly solved by every method; M&A and absorption events are where the v10 ensemble's gains over v2 are most concentrated (+9 pp and +12 pp respectively).

On the held-out-time split (Figure 4), the cross-encoder's gain is concentrated in the operationally hard events. Pure renames (`CHANGE_LEGAL_NAME`, $n = 10$) are essentially solved by every method including zero-shot E5-small (0.90 Hits@1, no lift). The headline gains come from `MERGERS_AND_ACQUISITIONS` (n = 1,506: 0.18 zero-shot → 0.62 v2 → 0.71 v10) and `ABSORPTION` ($n = 461$: 0.23 → 0.53 → 0.65), the cases where the post-event name comes from the acquirer rather than the target and pure name similarity is structurally unable to bridge the gap. v10 also more than doubles zero-shot on the small but operationally critical `BREAKUP` subset ($n = 7$: 0.14 → 0.71).

On the held-out-event split, "held-out event type" really means "held-out absorption." The held-out-event test set is 94.5% `ABSORPTION` anchors (Section 3.4); `DEMERGER`, `BREAKUP`, and `SPINOFF` contribute $n = 42$, $n = 20$, and $n = 4$ respectively. We report Table 3 so that the contribution of each held-out type is visible separately. The cross-encoder lift is consistent across the absorption-like types (`ABSORPTION`: 0.22 → 0.57, `DEMERGER`: 0.29 → 0.55, `BREAKUP`: 0.15 → 0.40); the `SPINOFF` cell is statistically meaningless at $n = 4$. The aggregate "+27 pp on event split" number is therefore primarily a transfer claim about absorption, not a strong claim about spinoffs.

5.6 Robustness to candidate-pool shift

The headline cross-encoder result is produced and (mostly) evaluated against candidates drawn from the same family of hard-negative miners — v1 token-overlap and v2 E5-cosine both rank candidates by name-related signals, and the cross-encoder is trained on v2 negs. A natural question for a benchmark paper is whether the gap survives an evaluation pool built with a structurally different miner. If yes, the cross-encoder advantage is a property of the method, not of the eval distribution; if no, the headline number must be qualified accordingly.

Table 3: Per-anchor-event-type Hits@1 on the held-out-EVENT test split, for the zero-shot baseline (E5-small) and the v2 cross-encoder. The test split is 94.5% `ABSORPTION`; `SPINOFF` at $n = 4$ is statistically uninformative. Lift sources are concentrated in the absorption-like types.

Event type	n anchors	zero-shot E5	v2 cross-enc	Δ
ABSORPTION	1,546	0.223	**0.568**	+34.5 pp
DEMERGER	42	0.286	**0.548**	+26.2 pp
BREAKUP	20	0.150	**0.400**	+25.0 pp
SPINOFF	4	0.500	0.250	−25.0 pp[†]

[†] $n = 4$, statistically uninformative.

Setup. We re-mine hard negatives with sparse char-3gram TF-IDF cosine (same signal as the `tfidf_char` baseline) — chosen because it has no learned representation and shares no parameters with E5. The other three classes (parent/subsidiary, same-name same/diff-country) are held constant. We call the resulting pair set TemporalLink-v3; splits and positives are unchanged. We train two cross-encoders (one on v2 hard negs, one on v3) and score each on both pair sets. The 2×2 train/evaluate matrix has diagonal cells in-distribution, off-diagonal cells out-of-distribution at the candidate-pool level.

Three findings. First, the cross-encoder *architecture* generalizes well across miners: the v3-trained cross-encoder evaluated on the v2 pool reaches 0.588 / 0.533 / 0.750 Hits@1 — within 1–4 pp of the v2-trained model on the same pool, despite never having seen E5-mined candidates during training. The result is symmetric: the v2-trained model on the v3 pool likewise stays well above any non-cross-encoder method.

Second, there is a real but bounded in-distribution effect. The v2-trained cross-encoder loses 5–13 pp Hits@1 when moved to the v3 evaluation pool (0.600 → 0.545 on time; 0.705 → 0.571 on jurisdiction). Conversely, the v3-trained model gains +7 pp on v3 over the v2-trained one. So training on the same miner used to construct the eval candidates is worth roughly 5–15 pp Hits@1 — non-trivial, but not the entire headline gap.

Third, the cross-encoder gain over zero-shot E5 *is robust* to the miner if we are willing to retrain the cross-encoder: the diagonal cells (in-distribution) show +20 to +40 pp gains across all six splits. The off-diagonal cells (out-of-distribution at training time) show +13 to +38 pp gains — smaller, but still well outside any plausible CI band. We read this as: joint encoding is structurally helpful for lifecycle-aware ER, but the absolute headline number is partly a function of how the evaluation candidate pool is constructed, and the "+24–35 pp" figure in the abstract should be qualified accordingly.

Fielded retrieve-then-rerank. The miner-robustness experiment above shifts the mining signal but still draws candidates from a curated negative pool. A real deployment pipeline does something different: a first-stage retriever pulls the top-K candidates from the full entity universe, then a second-stage model reranks. We simulate that here. The trained `method_v2_full` bi-encoder retrieves top-20 candidates per test anchor from the full 3.3M-entity GLEIF universe, restricted to the anchor's jurisdiction (the lightest filter a production system would apply); the cross-encoder reranks.

Bi-encoder Recall@20 is 33.2 / 39.6 / 49.7% on TIME / EVENT / JURIS. That is the ceiling for any reranker. When the bi-encoder does retrieve the true successor, the v2 cross-encoder picks it #1 with 46–58% Hits@1 and the v10 ensemble reaches 78–91% — the reranker is the strong link. End-to-end deployment Hits@1 (recall × conditional, with un-retrieved truths counted as misses) is 15–29% for v2 and 26–45% for v10. The drop from 60–75% in-distribution to 15–45% deployment confirms rather than contradicts §4.1: the bi-encoder's fixed-embedding limit, which fails to retrieve lifecycle-spanning successors by cosine similarity, is exactly the bottleneck the anti-correlation analysis predicts.

5.7 Ablations

Both ablations are in Table 1.

State features help modestly. Removing jurisdiction and legal form from the prompt (`method_no_state`) costs 1.5–2 pp Hits@1 vs. `method_full` on every split. The biggest contribution is on held-out-jurisdiction (where the country token does most of the work) and the smallest is on held-out-event-type (where the name change itself does it).

Explicit hard negatives slightly *hurt* in v1. `method_no_hard` edges out `method_full` by 0.4–1.1 pp Hits@1 on every split. The culprit is the `token_overlap_same_country` class, 92% of all mined v1 hard negatives and the lowest-precision class — the fuzzy fallback injects label noise, and the in-batch negatives from MultipleNegativesRanking are cleaner on their own. This motivates the v2 mining switch from token-overlap to E5-cosine, and pushes back on the default assumption that more hard negatives is always better.

5.8 Additional diagnostics

Four diagnostics, compressed to fit the page budget; full tables and figures are in the released code. **B-cubed cluster F1** [1] on the test partition is 0.92 / 0.90 / 0.91 for the cross-encoder vs. ≤0.47 for any bi-encoder (each of which collapses to one giant cluster at the tuned threshold). Restricting to records in true clusters of size ≥ 2 — 64% of test records are singletons — gives a more honest 0.85 / 0.84 / 0.88. **Calibration and selective prediction.** ECE (10-bin) is 0.20, Brier 0.18: the model is over-confident in its high-score bin, consistent with recent PLM-EM calibration results [10]. A selective-prediction [5] sweep on v10 hits Hits@1 ≥ 0.94 at 20% coverage and 0.87–0.97 at 50% across the three splits; users who need high precision can just threshold on the score (curve in repo). **Cross-domain transfer.** Zero-shot Magellan Walmart-Amazon: the cross-encoder *loses* to its own zero-shot base encoder by 5.9 pp AUROC. The fine-tuned representation is specialised to lifecycle-spanning legal-entity matching, not generic ER. **Secondary label-quality check.** 277 sampled positives re-classified by an orthogonal heuristic agree with the primary GLEIF auto-classifier on 98.7% [95% CI 97.0–100%] of the confident (231/277) subset, bounding label noise below 2% there; the other 16.6% were marked uncertain.

Table 4: Hits@1 with 95% bootstrap CIs for the 2×2 train/evaluation matrix. "v2 pool" refers to `pairs.parquet` (v1 token-overlap fallback, dominant negative class); "v3 pool" refers to `pairs_v3.parquet` (TF-IDF char-3gram cosine, no neural representation). E5-small zero-shot is included as a reference baseline on each pool. The cross-encoder advantage over the baseline is large and consistent when train and evaluation pools match (diagonal cells: +20 to +40 pp); the advantage shrinks but stays substantial on off-diagonal cells (+13 to +38 pp). The numbers reproduce: the original-paper `method_v2_crossenc` predictions and the re-trained `trn-on-v2` cells agree within 2 pp on time/event splits.

	Eval on v2 pool (`pairs.parquet`)			Eval on v3 pool (`pairs_v3.parquet`)		
	TIME	EVENT	JURIS	TIME	EVENT	JURIS
E5-small zero-shot	0.203	0.234	0.432	0.265	0.319	0.441
	±1.8	±2.1	±10.5	±1.4	±1.7	±7.8
cross-enc trn-on-v2	**0.600**	**0.551**	0.705	0.545	0.528	0.571
	±2.2	±2.4	±8.8	±1.8	±1.7	±7.5
cross-enc trn-on-v3	0.588	0.533	**0.750**	**0.655**	**0.614**	**0.640**
	±2.0	±2.4	±9.1	±1.5	±1.7	±7.1
Δ *vs. E5, in-distribution*	+39.7	+31.7	+27.3	+39.1	+29.7	+19.9
Δ *vs. E5, out-of-distribution*	+38.6	+29.9	+31.8	+28.0	+20.9	+13.0

5.9 Where each method is fooled

The negative-class breakdown (Table 5) reports per- negative-class mean score: higher means the method is more often fooled by that class of negative. The bi-encoder fine-tunes roughly halve the zero-shot E5 baseline's score on `parent_subsidiary` and `same_name_diff_country` negatives, but `same_name_same_country` remains a wall at ≈0.90 mean score: the bi-encoder cannot reliably tell a real successor link from two distinct entities sharing an exact normalized name in the same country.

The cross-encoder breaks that wall: `same_name_same_country` mean score drops from 0.90 (best bi-encoder) to 0.45, and the v10 ensemble drops it further to 0.34. v10 also nearly solves `same_name_diff_country` (0.92 zero-shot → 0.16) and `token_overlap_same_country` (0.91 → 0.25). The one regression is `parent_subsidiary` (0.45 bi-encoder → 0.70 v2 cross-encoder → 0.78 v10): the joint encoder occasionally treats parent–subsidiary edges as positive lifecycle bridges, and the v10 prompt enrichment that injects GLEIF L2 parent/child names makes the regression *worse*, not better. With only 221 examples in this class the headline impact is small, but the trend points at a clean v3 target: explicit multi-class modeling of corporate-hierarchy relationships.

Table 5: Negative-class mean score (lower is better). *p/s* = parent/subsidiary, *sn-D* / *sn-S* = same-name diff / same-country, *tok* = token-overlap same-country. v10 improves three classes but *worsens p/s* (0.70 → 0.78) — prompt-injected L2 context makes parent/subsidiary pairs look more similar, not less.

Method	p/s	sn-D	sn-S	tok
`exact_name`	0.013	1.000	1.000	0.000
`token_set`	0.144	1.000	1.000	0.486
`tfidf_char`	0.227	0.630	0.804	0.590
`sentence_encoder`	0.834	0.921	0.963	0.905
`method_full`	0.454	0.454	0.899	0.736
`method_v2_full`	0.457	0.464	0.899	0.736
`method_v2_crossenc`	0.696	0.237	0.452	0.278
`v10 ensemble`	0.781	**0.160**	**0.345**	**0.254**

6 ANALYSIS

6.1 Error analysis

We sampled 100 `method_full` failures on the held-out-time test split (positives that were not ranked first against their hard-negative set) and categorized them via simple text features. The categories:

Category	Count
other (mixed)	66
anchor and true successor share *no* tokens	19
substring name overlap with a wrong candidate	8
exact normalized-name collision with a wrong candidate	7

The 19 "no shared tokens" cases are the hardest: a name like `Nima N.V.` (NL) whose true successor is `Beheermaatschappij Arros B.V.` shares zero meaningful tokens with either the predecessor or the wrong top-1 candidate `NIMA B.V.`. No similarity-based representation can solve these; they require world knowledge or explicit graph context (board overlap, regulatory filings, press releases). One representative failure: anchor `DB HR Solutions GmbH` (DE), true successor `DEUTSCHE BANK AKTIENGESELLSCHAFT`, top-1 wrong `HR Solutions GmbH` (a token-overlap negative). The DB→Deutsche Bank disambiguation requires recognizing "DB" as a known abbreviation for "Deutsche Bank" — an external knowledge fact a 33M-parameter encoder is unlikely to memorize from name strings alone.

v10 residual JURIS failures cluster on sibling consolidation. Of the 20 / 161 anchors v10 mis-ranks on JURIS, ≥11 are sibling consolidations (e.g., 9 "`EPIC <port> PTE. LTD.`" Singapore ships into `BW EPIC KOSAN VESSELS ONE`; the model returns sibling `EPIC` vessels); a second cluster is Japanese-script name changes the ASCII tokenizer cannot bridge (Daiwa-Sumigin → Mitsui-Sumitomo DS). An LLM with multilingual pretraining and ownership-graph context would address both.

6.2 Why this matters in practice

The benchmark targets a workflow primitive shared by every fintech infrastructure company we are aware of: Stripe trains merchant embeddings [13] and fights AI-driven rebranding fraud [14]; Ramp performs vendor deduplication across ∼27M spreadsheet cells in its agentic-accounting product [12]; Bloomberg's entity backbone underpins both its NLP work and the Quant Research Delphyne foundation model [2]; Robinhood's Cortex assistant reconciles tickers, CIKs, and corporate-action histories across pre- and post-event references. Each team reports lifecycle-ER failures as recurring operational pain points; TemporalLink gives them a public benchmark to evaluate against.

7 LIMITATIONS AND FUTURE WORK

Six limitations; the released code carries the longer versions (cost/latency, calibration curves, error taxonomy, label-quality).

The "anti-correlation" finding is a construction artefact. Hard negatives in TemporalLink are mined to be name-similar to the predecessor; positives are by definition name-different. The sub-0.5 AUROC of similarity baselines (Section 5) follows directly from that mining design and does not generalise to ER outside this benchmark. The finding motivates joint encoding here; it is not a universal claim about name similarity in ER.

In-distribution headline overstates deployment. The headline cross-encoder Hits@1 (60–75 %) is on the v2 mined candidate pool; on a fielded retrieve-then-rerank pipeline that draws candidates from the full 3.3M-entity GLEIF universe (§5.6), end-to-end Hits@1 falls to 15–29 % (v2) / 26–45 % (v10). The cause is upstream: bi-encoder Recall@20 is 33–50 %, so the reranker often never sees the true successor. The miner-robustness experiment shows the cross-encoder gain itself is partly in-distribution but not purely a within-miner artefact (+13 to +38 pp off-diagonal). A stronger first-stage retriever — a learned hybrid sparse-dense index, or a coarsely pre-filtered cross-encoder shortlist — is the highest-leverage v3 target.

Seed variance and a small hard-negative label noise. Bi-encoder Hits@1 over seeds 42/7/1337 was 0.251 ± 0.005 (TIME), 0.291 ± 0.004 (EVENT), 0.523 ± 0.016 (JURIS); TIME/EVENT are stable, JURIS varies ±1.6 pp consistent with its small test sample ($n = 88$). Cross-encoder multi-seed retraining (v4 ensemble, Table 2) shows similar behaviour. Separately, 205/93,695 = 0.22% of hard negatives appear in GLEIF's `SuccessorEntity`; the cleaned variant `pairs_clean.parquet` drops them with ≤0.2 pp impact.

Reporting / survivorship bias + thin slices. GLEIF `SuccessorEntity` is voluntarily populated, biasing positives toward well-counseled regulated entities. The held-out-event test set is 94.5% absorption (the label oversells event-type generality), and the held-out-jurisdiction test has only 88 anchors (±8.8 pp Hits@1 CI). The cross-encoder gain clears the CI band on all three splits, but per-split absolute numbers should be read with those sample sizes in mind. A secondary auto-classifier check (Section 5.8) bounds label noise at <2% on the confident-positive subset; expert multi-annotator review would give a tighter bound.

LLM baselines are sampled, not full test set. Two LLM rows in Table 1 are evaluated on a stratified test-set sample (56 / 52 / 50 anchors for TIME / EVENT / JURIS), with each anchor's full candidate pool scored. Hits@1 95% bootstrap CIs: Haiku 4.5 ±11/13/14 pp on TIME/EVENT/JURIS; gpt-oss-20b ±10/11/14 pp. Both LLMs trail the fine-tuned cross-encoder by 30–40 pp on every split — a gap larger than the LLM-row CI bands. We did not run a Sonnet / Opus / GPT-4o-class upper-bound model: an in-progress qwen3-32b run was abandoned when Apple-Silicon throughput proved impractical (>3 h wall-time for the 1,165 sampled pairs even with thinking disabled). Running the closed-API panel on the full benchmark (≈$200) and adding a frontier model would tighten the upper-bound estimate; we leave that to a revision with a budget.

Cross-domain transferability + small-n caveat on parent/subsidiary. The cross-encoder underperforms zero-shot E5-small on out-of-domain product matching (Walmart-Amazon, −5.9 pp AUROC; Section 5.8); the fine-tuned representation is specialized to lifecycle ER, not generic. The parent/subsidiary negative class shows a regression (mean score 0.45 → 0.70 from bi-encoder to v2 cross-encoder, worsening to 0.78 in v10), but with only 221 negatives the effect-size estimate is not statistically meaningful. Explicit multi-class modeling of related-but-distinct entities is a v3 target.

8 CONCLUSION

TemporalLink is an open benchmark for lifecycle-aware entity resolution on financial entities, built from public GLEIF, SEC EDGAR, and Wikidata data and reproducible end-to-end on a single MacBook in ∼2 hours. Hard negatives are mined to be name-similar to each predecessor and true successors are by construction name-different, so similarity-based baselines fall below 0.5 AUROC on the time and event splits — a property of this benchmark's construction, not of name similarity in ER generally. A 22M-parameter MiniLM cross-encoder is the first method here to cross 0.5 AUROC, reaching 60–75% Hits@1 and sitting 24–35 pp above the strongest bi-encoder; paired McNemar tests put the gap over a Claude-Haiku-4.5 LLM baseline at $p < 10^{-3}$ on every split. A miner-robustness re-mine with TF-IDF char-3gram cosine shrinks the cross-encoder advantage 5–15 pp but keeps it +13 to +38 pp above zero-shot, and a prompt-and-backbone extension lifts the headline a further +9 to +11 pp. A fielded retrieve-then-rerank simulation over the 3.3M-entity universe drops end-to-end Hits@1 to 15–29% (v2) / 26–45% (v10) because the bi-encoder's Recall@20 caps at 33–50%; the cross-encoder reranker remains strong. Per-event analysis is the most honest summary: real gains on M&A, absorption, and breakup events; pure renames are already at 0.90 Hits@1 zero-shot. The next iteration's biggest lever is a stronger first-stage retriever; smaller targets include a full-test-set LLM panel, an expanded held-out-jurisdiction set, and multi-class modelling of related-but-distinct entities.

Reproducibility. All code, parsers, prediction artifacts, trained model weights, and the miner-robustness experiment are released at https://github.com/chang-07/er-research. End-to-end pipeline reproduction from the raw GLEIF XML — benchmark, both

methods, v3 robustness, and the v10 ensemble — takes ~2 hours on a single MacBook laptop.

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